Resin Slicing & Printing Guide

Getting Started:

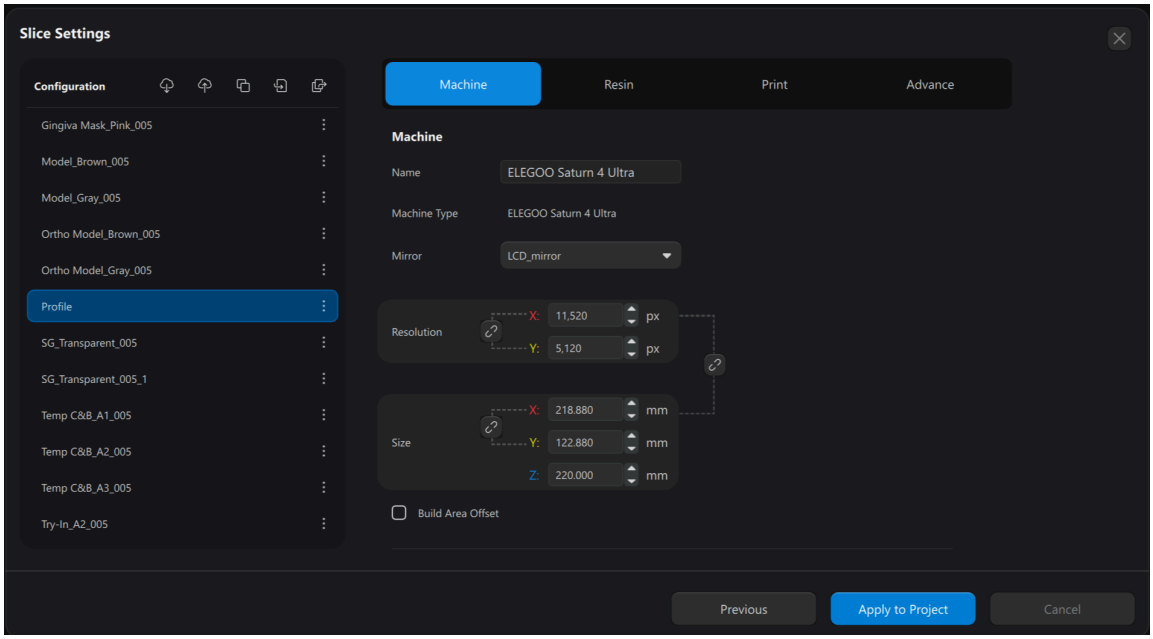
- Navigate to https://www.chitubox.com/en/index and download the CHITUBOX slicer software
- Follow the setup instructions

Creating Slicing Profiles:

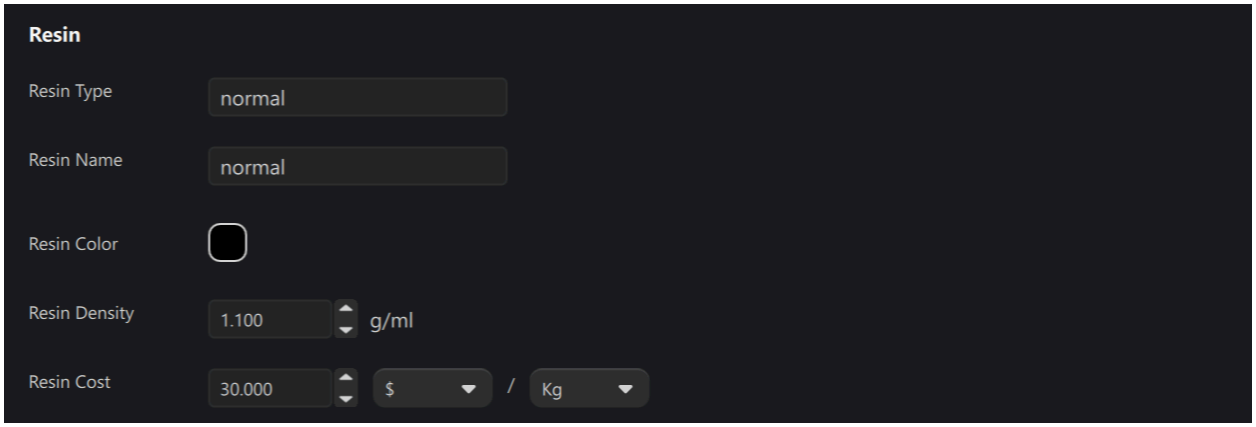
- Select the ELEGOO Saturn 4 Ultra & ELEGOO Jupiter SE Printers



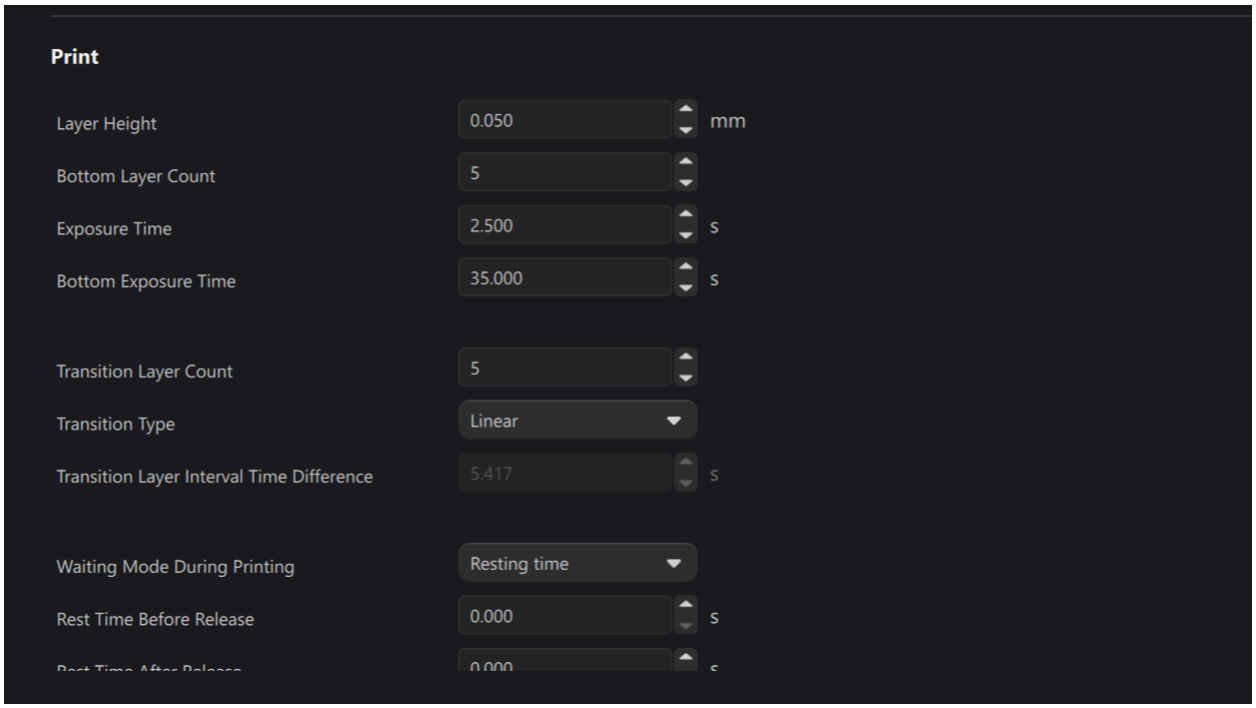
- Ensure you are under the “Profile” Configuration as seen on the left



- Scroll down to the “Resin” section and match the resin color and density to match your desired resin.
- **NOTE:** Changing these settings will not alter your actual print, and will only affect the weight and cost measurements to slicer automatically performs



- Scroll down to the “Print” Section
- Based on manufacturer specifications, alter the “Exposure Time” and “Bottom Exposure Time”
- **NOTE:** It is not necessary to change “Layer Height” and “Bottom Layer Count”. However, altering these parameters could be done in order to achieve even greater print quality.



Determining Exposure Times:

- Some manufacturers do not have the capabilities to test every resin on every printer, so it is up to you (or other people online) to determine the best cure time per layer for each printer/resin.
- From experience, the prints on the Jupiter SE should have an "Exposure Time" increase of 0.5s more than the Saturn 4 Ultra.

Quantitative Method

- Slice 3-5 Dogbones at various exposures times (ex. 2.5s, 3s, 3.5s) at a 45 degree angle.
- Wash and post-cure the dogbones how you typically would a resin part
- Contact gr-mmeteachinglab@wpi.edu to use the Mecmesin MultiTester. This is a tensile tester that can be used to pull your samples apart.
- Determine what exposure time per layer samples had the highest Young's Modulus by comparing the stress vs. strain data.
- To further refine exposure time, print more samples at exposure times between your two best performing exposure times per layer batches.

Qualitative Method

- Navigate to websites such as https://www.thingiverse.com/ and download a resin printing exposure test. These are designs that can be used to qualitatively determine whether your exposure times are too high or low.
- Furthermore, utilizing prints that measure resin expansion can be helping in determining if you need to use tolerance compensation in the CHITUBOX slicer.

Slicing Basics